```
> restart;
    # h Planck , audion, monopole microwaves,monopole Radio Frequency
> h := 6.6260755·10^-34; c := 299792458; hbar := 6.62607015/(2·Pi)·10^-34; Tq := 1.76536
    ·10^-19; eq := 1.602176634×10^-19;
```

$$h := 6.626075500 \times 10^{-34}$$

$$c := 299792458$$

$$\hbar := 1.054571818 \times 10^{-34}$$

$$Tq := 1.765360000 \times 10^{-19}$$

$$eq := 1.602176634 \times 10^{-19}$$

$$h\gamma := 9.340461794 \times 10^{-18} \quad (1)$$

```
> he := 4.135667697×10^-15; #h Planck in eV; hme := 5.663809031
    ×10^-9 the magnetic minimum action eV; hAe :=
    0.007481060606 costant of minimum action of Audion in eV;
```

$$he := 4.135667697 \times 10^{-15} \quad (2)$$

```
> solve(hγ·c/(7.8·10^(-10)) = 3.59); 9.340461794×10^-18·c/54; c/(0.518·10^(-10))·9.340461794
    ×10^-18
```

$$5.185555556 \times 10^{-11}$$

$$54.05791504 \quad (3)$$

```
> E01 := 4.135667697×10^-15·c/(380.4·10^-9);
```

$$E01 := 3.259311210 \quad (4)$$

```
>
> Et := E01 - 4.214492693×10^-38·c^2/(3.08^2·10^-18)
```

$$Et := 3.258911923 \quad (5)$$

```
> h0 := 4.214492693×10^-38;#the value of h0 in eV;
```

$$h0 := 4.214492693 \times 10^{-38} \quad (6)$$

```
>
> hme := 3.259054187/(575.4173859×10^6);# hm in eV;
```

$$hme := 5.663809031 \times 10^{-9} \quad (7)$$

```
> hm := hme·eq;# hm in Joules;
```

$$hm := 9.074422489 \times 10^{-28} \quad (8)$$

```
> solve(3.59 = hme·25×10^9 - hm1·25.095^2×10^18);
```

$$2.191397349 \times 10^{-19} \quad (9)$$

```
> hm1 := 2.208074846×10^-18;
```

$$hm1 := 2.208074846 \times 10^{-18} \quad (10)$$

```
>
> hm1 := 2.208074846×10^-18;
```

$$hm1 := 2.208074846 \times 10^{-18} \quad (11)$$

```
>
> EionizG := h·c/(380.43·10^-9);#Energy in J
```

(12)

$$EionizG := 5.221584683 \times 10^{-19} \quad (12)$$

```
> EionizG/eq
```

$$3.259056818 \quad (13)$$

```
> EionizGeV := he·c/380.43·10^-9;#Energy Gravtational Ionization 1 in eV;
```

$$EionizGeV := 3.259054186 \quad (14)$$

```
>
> he·c/313.5·10^-9 ; he·c/345.35·10^-9;
```

$$3.954838864$$
$$3.590102748 \quad (15)$$

```
> 0.188977/2.585/10^(-8);0.188977/1.057000000/10^(9);
  0.188977/1.787861873/10^(-10); 3.259/1.787861873/10^(-10);
  3.59/1.787861873/10^(-10); 3.95/1.787861873/10^(-10); 3.59/2.585/10^
  (-8);
```

$$7.310522244 \times 10^{6}$$
$$1.787861873 \times 10^{-10}$$
$$1.057000000 \times 10^{9}$$
$$1.822847754 \times 10^{10}$$
$$2.007985099 \times 10^{10}$$
$$2.209342936 \times 10^{10}$$
$$1.388781431 \times 10^{8} \quad (16)$$

```
> 3.259416157*380.43/3.954838513;3.590102429*345.35/3.954838513
```

$$313.5348472$$
$$313.5000000 \quad (17)$$

```
> EdegenSeV := he·c/345.35·10^-9;# Electron spin energy in eV
```

$$EdegenSeV := 3.590102748 \quad (18)$$

```
> EdegenSeV· eq;# Electron spin energy in J
```

$$5.751978737 \times 10^{-19} \quad (19)$$

```
> t := he/2·Pi·2· 2· EdegenSeV; # the time of electron trembling
```

$$t := 4.583517542 \times 10^{-17} \quad (20)$$

```
> (t/1.765·10^-19)^(-1)
```

$$0.003850754325 \quad (21)$$

```
> hbar/3.38·10^3· Tq;
```

$$1.767364710 \times 10^{-19} \quad (22)$$

```
> hme := 5.663809031 × 10^-9
```

$$hme := 5.663809031 \times 10^{-9} \quad (23)$$

```
> 3.259/hme;#The Frequency of gravitational ionisation =575.4 Mhz
```

$$5.754078185 \times 10^{8} \quad (24)$$

```
> 5.3193/hme;# The frequency
```

$$9.391736143 \times 10^{8} \tag{25}$$

```
> (3.259053897·3.955)^0.5;
```

$$3.590203081 \tag{26}$$

```
> h·c/(380.43·10^-9·575.4173859×10^6·1.602·10^-19);# hme = 5.664438088×10^-9 eV·s
```

$$5.664438088 \times 10^{-9} \tag{27}$$

```
> h*c/(1.602*((345.35*10^(-9))*6.447149633*10^8)*10^(-19))
```

$$5.569130067 \times 10^{-9} \tag{28}$$

```
>
> hAe := 0.007486790719; hAe := 3.259/435.3;
    # The costant of minimum action of Audion in eV;
```

$$hAe := 0.007486790719$$

$$hAe := 0.007486790719 \tag{29}$$

```
> hme/he; hAe/he; 181.4·10^6·380.4/3.08
```

$$1.369502931 \times 10^{6}$$

$$1.810297942 \times 10^{12}$$

$$2.240407792 \times 10^{10} \tag{30}$$

```
> 3.954838513/3.259416157·435.156; 3.954838513/3.590102429·435.156;
```

$$527.9999930$$

$$479.3656286 \tag{31}$$

```
> hAe/he; 3.95/hAe; 3.259/hAe;#The resonance frequency graviton phonon 435.4 Hz
```

$$1.810297942 \times 10^{12}$$

$$527.5958882$$

$$435.2999999 \tag{32}$$

```
> hme := 5.664438088×10^-9;
    #The Planck's ct of minimum action of monopole Radio Frequency in
    eV
```

$$hme := 5.664438088 \times 10^{-9} \tag{33}$$

```
> 22.48786848/hme;12.64/hme;# CMB for magnetron ~3.97÷2.234Mhz
```

$$3.970449630 \times 10^{9}$$

$$2.231713663 \times 10^{9} \tag{34}$$

```
> 528/3.955·3.259; 528·380.43/3.08;# Gravitonic Ionised State 435Hz & 65.2 Khz
```

$$435.0826800$$

$$65216.57143 \tag{35}$$

```
> 380.43·528/3.08;#65216·575×10^6/435.156;
```

$$65216.57143 \tag{36}$$

```
>
```

```
> 160.23*10^9·hmw ; 285·10^9·hmw;
   #Penzias and Wilson originally detected the Cosmic Microwave
   Background 160.23÷285 GHz with an energy of ~12.64÷22.49 eV ;
```

$$1.6023000000000 \times 10^{11}\ hmw$$

$$285000000000\ hmw \tag{37}$$

```
> 160.23*10^9*hmw;#The energy of CMB
```

$$12.64291637 \tag{38}$$

```
> #  http://hyperphysics.phy-astr.gsu.edu/hbase/quantum/h21.html
> #10.9Ghz is 3.59eV on the microwave thus we can deduce the Planck's
  constatnt for miccrowave;
> E:=hmw*nu;
```

$$E := 7.890480167 \times 10^{-11}\ \nu \tag{39}$$

```
> hmw:=3.59/(10.9*10^9);
```

$$hmw := 3.293577982 \times 10^{-10} \tag{40}$$

```
> 1239.84/22.48786848; 1239.84/12.6429 ; hmw·2.395·10^9; hmw·160.23·10^9; hme·2.395·10^9;
    evalf(c/(2.395·10^9));
```

$$55.13372693$$
$$98.06610825$$
$$0.1889770000$$
$$12.64291637$$
$$13.56482263$$
$$0.1251743040 \tag{41}$$

```
> 1239.84/22.48786848; 1239.84/12.64829;# The CMB photon UV radiation ~55.13÷98 nm
```

$$55.13372693$$
$$98.02431791 \tag{42}$$

```
> c/(160.23·10^9); c/(2.385·10^9);# The CMB Microwaves second value ~1.87mm
```

$$0.001871013281$$
$$0.1256991438 \tag{43}$$

```
> evalf(c/(285·10^9));#The CMB Microwaves second value ~1.05mm thus 285Ghz
```

$$0.001051903361 \tag{44}$$

```
> 7.8·3.59/22.48786848; 7.8·3.59/12.648786848# γ− ray of 1.2−2.4 Angstrom
```

$$1.245204721$$
$$2.213809145 \tag{45}$$

```
> 22.48786848/hme;12.64/hme;# CMB for magnetron ~2.23-3.97Ghz
```

$$3.970449630 \times 10^{9}$$

$$2.231713663 \times 10^{9} \tag{46}$$

```
> 22.48786848/hAe;12.64/hAe;# CMB for phonon ~1.7-3Khz
```

$$3003.672644$$
$$1688.306842 \tag{47}$$

```
> he;hmw;hme;hAe;`h&gamma;`
```

$$4.135667697 \times 10^{-15}$$

$$hmw$$

$$5.663809031 \times 10^{-9}$$

$$0.007486790719$$

$$9.340461794 \times 10^{-18} \quad \textbf{(48)}$$

```
> hγ := solve( k·c/(7.8·10^(-10)) = 3.59 );#Planck constant for γ-ray
```

$$h\gamma := 9.340461794 \times 10^{-18} \quad \textbf{(49)}$$

```
> `h&gamma;`:= 9.340461794*10^(-18);
```

$$h\gamma := 9.340461794 \times 10^{-18} \quad \textbf{(50)}$$